S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026 | DOSELECT PHASE III

FinRisk Intelligence

AI-Powered Financial Risk Engine & Strategic Multi-Asset Portfolio Stress Testing

Selected Downstream Track: Module B — Portfolio Stress Testing

Candidate & Team Identification

Candidate Name:	Ramisetty Aparanjith
College / University:	IIT (ISM) Dhanbad
Official Email:	23je0787@iitism.ac.in
Degree & Department:	B.Tech / Computer Science and Engineering
Public GitHub Repository:	https://github.com/Aparanjith25/FinRisk-Intelligence
Unlisted Demo Video:	https://youtu.be/hfqHQLqod7w

Technical Submission Highlights

- End-to-End Operational Pipeline: Unstructured narrative ingestion to multi-asset valuation.
- Genuine Dual-Source Ingestion: Live regulatory news (SEC, ECB, Fed) + synthetic social CSV.
- Financial NLP Risk Scoring: ProsusAI/finbert continuous sentiment [-1, 1] + 6 event categories.
- Explainable Impact Scoring: Transparent 1-10 integer score with additive audit rationales.
- High-Impact Event Alerts: Configurable threshold (> 7) with human-confirmed scenario staging.
- Module B Wholesale Portfolio: 10 assets ($17.0M baseline) including Pay-Fixed IRS with DV01.
- Engineering Rigor: 147 passed tests (0 failed, 0 skipped), FastAPI REST + Streamlit UI.

Problem Statement & Technical Approach

Financial Risk Industry Challenges

Deluge of Unstructured Text:
Risk analysts must continuously monitor central bank speeches, SEC filings, and fast-moving market narratives. Ma

Opaque Black-Box NLP:
Off-the-shelf generative models hallucinate entities, confuse publication dates, and provide unexplainable predicti

Lagging Risk Indicators:
Traditional VaR models rely on historical covariance matrices that lag during sudden macroeconomic regime shifts

Disconnection Between News & Portfolios:
Existing systems either generate news sentiment without portfolio links or run stress tests without narrative contex

Uncontrolled Execution Risk:
Automated execution of algorithmic trades or shock parameters without human validation introduces systemic risk

FinRisk Intelligence Approach & Principles

Multi-Stream Ingestion & Normalization:
Ingests live RSS feeds (Fed, SEC, ECB) alongside synthetic social streams into a canonical 7-field schema with determinist

Continuous Sentiment in [-1.0, +1.0]:
Computes P(pos) - P(neg) via ProsusAI/finbert with graceful offline deterministic baseline fallback and strict metadata at

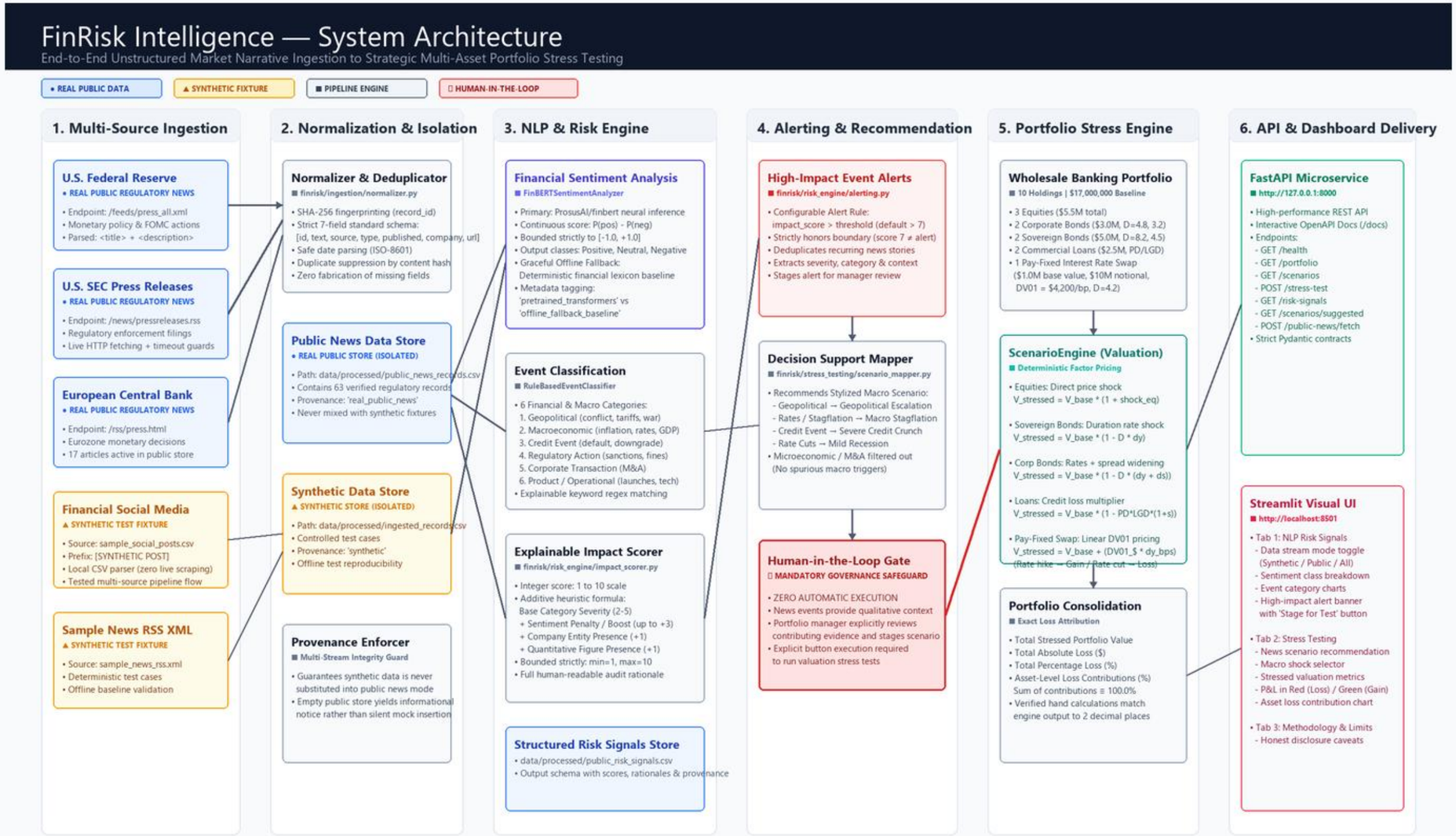
Explainable Rule-Based Impact Scoring:
Combines event categorization (6 types) with an additive 1-10 impact scoring heuristic. Every score emits a human-read

Human-in-the-Loop Decision Support:
High-impact events (> 7) trigger alerts that stage recommended macro scenarios. ZERO automatic executions; the port

Module B Quantitative Portfolio Valuation:
Simulates multi-factor shocks on a $17.0M wholesale portfolio (Equities, Corp/Govt Bonds, Loans, and an Interest Rate S

Confidential — Hackathon Submission

System Architecture & End-to-End Pipeline



Strict Architectural Separation: Real regulatory news (Fed, SEC, ECB) is isolated from synthetic test fixtures. All outputs flow to FastAPI and Streamlit.

Implementation Details & Technology Stack

1. Ingestion & Normalization Layer

Dual Stream Parsers:
XML/RSS parser for news + CSV tabular parser for social feeds.

Public Syndication Feeds:
Direct live fetching from SEC and ECB RSS feeds (10s bounded timeout).

SHA-256 Fingerprinting:
Deterministic record IDs eliminate duplicate stories across multiple runs.

Zero Hallucination:
Missing company names or publication dates are preserved as None, never fabricated.

Provenance Isolation:
Real public records and synthetic records are stored in isolated file paths.

2. NLP & Explainable Risk Scoring

Sentiment Engine:
ProsusAI/finbert yielding continuous P(pos) - P(neg) in [-1.0, +1.0].

Offline Fallback:
Deterministic domain lexicon fallback when torch/transformers are omitted.

Event Categorization:
6 categories: Geopolitical, Macroeconomic, Credit, Regulatory, M&A, Product.

Impact Scoring Heuristic:
Additive integer score (1-10) factoring base severity, sentiment, and metrics.

Audit Trail:
Every risk signal generates a transparent, human-readable rationale explanation.

3. Stress Testing Valuation (Module B)

Multi-Asset Coverage:
Equities, Corporate Bonds, Sovereign Bonds, Loans, and Interest Rate Swaps.

Derivative Support:
5Y Pay-Fixed Macro Hedge Swap ($10M notional, DV01 = $4,200/bp, D=4.2).

Directional Consistency:
Rate hikes produce swap gains; rate cuts produce swap losses.

Loss Attribution:
Asset-level contributions sum exactly to 100.0% of total portfolio loss.

Standardized Scenarios:
4 preconfigured macro shocks (Stagflation, Geopolitical, Credit, Recession).

4. API, Dashboard & Testing Stack

FastAPI Microservice:
High-performance async REST backend with OpenAPI Swagger docs (/docs).

Streamlit Analytics UI:
Interactive 3-tab dashboard with real-time feed switching and Plotly charts.

Robust Fallback Pattern:
Dashboard operates seamlessly with live API or direct local engine fallback.

Automated Test Suite:
147 unit, integration, and UI AppTests passing with 100% pass rate in ~41s.

Python Stack:
Built on Python 3.13, Pydantic, Pandas, NumPy, HTTPX, and Pytest.

Confidential — Hackathon Submission

Empirical Results & Validated Workflow

Automated Tests Passing

147 / 147

100% Pass Rate (0 failed, 0 skipped)

Real Public News Ingested

63 Records

26 SEC, 20 Fed, 17 ECB Press Releases

Portfolio Market Value

$17,000,000

10 Assets Across 5 Distinct Classes

Validated End-to-End Pipeline Trace (Real Data)

Step 1: Ingestion	63 genuine regulatory records loaded from data/processed/public_news_records.csv
Step 2: Normalization	100% verified with SHA-256 IDs, provenance='real_public_news'
Step 3: Sentiment Scoring	ProsusAI/finbert continuous scores range: [-0.9208, +0.9131] (strictly within [-1, 1])
Step 4: Event Classification	Macroeconomic (26), Regulatory (18), Geopolitical (1), Corporate (18)
Step 5: Impact Scoring	Impact scores range: [1, 7] strictly within [1, 10] with additive audit rationales
Step 6: High-Impact Alerting	threshold=6: 4 alerts detected and staged; at threshold=7: 0 alerts (boundary honored)
Step 7: Scenario Mapping	Signals recommended 'Macroeconomic Stagflation' (Composite score: 0.48, 12 triggers)
Step 8: Portfolio Stress Test	$17.0M portfolio stressed to $15,960,740.00 (-$1,039,260.00 / -6.11% total loss)
Step 9: Delivery	Results rendered in Streamlit UI & FastAPI (/stress-test) with holding-level waterfall

Empirical NLP Sentiment Evaluation

Preliminary Curated Regulatory Benchmark (N = 30):

- ProsusAI/finbert: 76.67% Accuracy | Macro F1: 0.7548
- Deterministic Baseline: 70.00% Accuracy | Macro F1: 0.6849
- Held-out ECB subset (n=8): Baseline 87.5% vs FinBERT 75.0% (due to hedged central-bank neutral wording)

Financial PhraseBank Evaluation (N = 339 test split):

- ProsusAI/finbert: 97.35% Accuracy | Macro F1: 0.9628
- Fine-tuned candidate on corporate news: 98.53% Accuracy | Macro F1: 0.9746

Advanced NLP Experimentation & Routing Finding:

- Fine-tuning on corporate news caused catastrophic forgetting on legal/regulatory text (regulatory negative recall dropped 70% → 60%).
- Production model strictly remains original ProsusAI/finbert.
- Built isolated source-aware routing prototype in experiments/; recommends more regulatory training data before production deployment.

Confidential — Hackathon Submission

Domain Impact & Module B Stress-Testing Case Study

Case Study: Ingested FOMC & Energy Shocks Triggering 'Macroeconomic Stagflation' Shock

Scenario Parameters: Equity -15% | Yield Curve +150 bps | Credit Spreads +100 bps | Commercial Loan Loss +20%

ID	Asset Name	Asset Class	Base Value	Stressed Value	Loss / Gain ($)	P&L %	Contribution	Valuation Method
EQ_001	AlphaCorp Common Stock	Equity	$2,500,000	$2,125,000	**-$375,000.00**	-15.00%	36.08%	Direct Price Shock (-15%)
EQ_002	BetaEnergy Global Shares	Equity	$1,800,000	$1,530,000	**-$270,000.00**	-15.00%	25.98%	Direct Price Shock (-15%)
EQ_003	GammaTech Solutions	Equity	$1,200,000	$1,020,000	**-$180,000.00**	-15.00%	17.32%	Direct Price Shock (-15%)
CB_001	GammaBank Senior Notes	Corp Bond	$2,000,000	$1,760,000	**-$240,000.00**	-12.00%	23.09%	Duration (D=4.8, dy+ds=250bp)
CB_002	DeltaMfg Credit Debentures	Corp Bond	$1,000,000	$920,000	**-$80,000.00**	-8.00%	7.70%	Duration (D=3.2, dy+ds=250bp)
GB_001	US Treasury Benchmark 10Y	Govt Bond	$3,000,000	$2,631,000	**-$369,000.00**	-12.30%	35.51%	Duration (D=8.2, dy=150bp)
GB_002	German Bund Sovereign 5Y	Govt Bond	$2,000,000	$1,865,000	**-$135,000.00**	-6.75%	12.99%	Duration (D=4.5, dy=150bp)
LN_001	CRE Syndicated Senior Loan	Loan	$1,500,000	$1,471,650	**-$28,350.00**	-1.89%	2.73%	Credit Migration (PD*LGD*1.20)
LN_002	SME Secured Term Facility	Loan	$1,000,000	$964,000	**-$36,000.00**	-3.60%	3.46%	Credit Migration (PD*LGD*1.20)
SW_001	5Y Pay-Fixed IRS Macro Hedge	Derivative	$1,000,000	$1,630,000	**+$630,000.00**	+63.00%	-60.62%	Linear DV01 ($4,200/bp * +150bp)

CONSOLIDATED PORTFOLIO STRESS RESULTS:

Baseline Market Value: $17,000,000.00 → Stressed Portfolio Value: $15,960,740.00

Total Absolute Loss: -$1,039,260.00 (-6.11%) | Total Holdings: 10 | Loss Attribution Check: Sum of Asset Losses = $1,039,260.00 (Exact Match)

DERIVATIVE MACRO HEDGE IMPACT:

The Pay-Fixed IRS gained +$630,000 (+63.0%), actively offsetting 37.7% of physical bond losses and cushioning overall portfolio drawdown.

Confidential — Hackathon Submission

Model Limitations, Honest AI Principles & Future Roadmap

Explicit Modeling Limitations (Non-Exaggerated)

Stylized Sensitivity vs. Econometric Forecasting:
Portfolio stress testing executes deterministic sensitivity shocks. It does NOT predict future market prices, default probabilities, or cal

Qualitative Decision Support vs. Quantitative Calibration:
News risk signals recommend stylized macro scenarios for human review; they do NOT automatically calibrate quantitative basis-po

Linear DV01 & Duration Approximations:
Fixed-income and swap revaluation utilizes first-order linear duration / DV01 without second-order convexity adjustments or yield c

Social Media Stream is Synthetic:
Ingested social media posts are local synthetic fixtures ([SYNTHETIC POST]), not a live Twitter/Reddit scraping stream (avoids enterp

RSS Summary vs. Full Text Scraping:
News ingestion parses official RSS <title> and <description> fields (~30-80 words) rather than scraping full HTML articles behind pu

Engineering Rigor & Post-Hackathon Roadmap

Source-Aware Sentiment Routing Prototype:
Deploy the experimentally validated routing prototype (using original FinBERT for regulatory text and fine-tuned models for corporate fili

Full Yield Curve Term Structure Repricing:
Integrate QuantLib or open-source curve bootstrapping to price non-parallel curve twists, steepening, and second-order swap convexity.

Live Enterprise Social & News Webhooks:
Connect official enterprise APIs (Bloomberg, FactSet, X v2 API) with rate-limit queuing and audit logging.

Counterparty Credit Valuation Adjustments (CVA/DVA):
Incorporate bilateral counterparty exposure profiling into derivative valuation.

Automated CI/CD Regression Pipeline:
Maintain the strict 147-test automated verification suite across all future PRs and model updates.

Confidential — Hackathon Submission